

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (IT)

Nov 2019

IT371/IT316 Advanced Java Programming

Date: 14.11.2019, Thursday

Time: 10:00 am To 01:00 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following questions:

[07]

a) Which of the following is used to call stored procedures on the database?

- | | |
|------------------------------|-------------------------------|
| A - Statement | B - PreparedStatement |
| C - CallableStatement | D - None of the above. |

b) Why will you set auto commit mode to false?

- | | |
|--|---|
| A - To increase performance. | B - To maintain the integrity of business processes. |
| C - To use distributed transactions | D - All of the above. |

c) Which of the following is a Metadata interfaces of JDBC?

- | | |
|-------------------------------|-------------------------------|
| A - DatabaseMetaData | B - ResultSetMetaData |
| C - Both of the above. | D - None of the above. |

d) Which class is used to create servers that listen for either local client or remote client programs?

- | | |
|--------------------------|-------------------------------|
| A - ServerSockets | B - httpServer |
| C - httpResponse | D - None of the above. |

e) Which classes are used for connection-less socket programming?

- | | |
|---------------------------|-------------------------------|
| A - DatagramSocket | B - DatagramPacket |
| C - Both A & B | D - None of the above. |

f) Which steps occur when establishing a TCP connection between two computers using sockets?

- A -** The server instantiates a ServerSocket object, denoting which port number communication is to occur on.
- B -** The server invokes the accept() method of the ServerSocket class. This method waits until a client connects to the server on the given port.
- C -** After the server is waiting, a client instantiates a Socket object, specifying the server name and port number to connect to.
- D -** All of the above

g) Which of these is a process of writing the state of an object to a byte stream?

A - Serialization

B - Externalization

C - File Filtering

D - All of the above.

Q - 2 (a) Differentiate execute, executeQuery, executeUpdate methods of JDBC. List the benefits of PreparedStatement over Statement. **[07]**

(b) What is datagram? What are some key classes defined in the Java programming language to work with datagrams? How do you get a list of IP addresses that are assigned to a network interface? **[07]**

OR

Q - 2 (a) What is ResultSet? Explain the different types of result set based on the concurrency and cursor types. **[07]**

(b) Differentiate Serializable and Externalizable interface in Java. How many methods are there in the Serializable? Why? **[07]**

Q - 3 (a) Write following java TCP socket programs. **[07]**

1) Server program: The program will accept the string from the client and it will check whether the string is palindrome or not. If it is palindrome then program will return "True" otherwise it will return "False"

2) Client program: The client program will accept string from the user and send it to the server. After receiving the server's response, the client program will print "The string is palindrome" or "The string is not palindrome" based on the server's response.

(b) Write the jdbc programs having following functionalities: **[07]**

1) Create a table "Employee" having a three fields: ID (Number), Name (Varchar), Mobile No (Number).

2) Input the data using prepared statement into the table.

3) Fetch and list the records of the table.

OR

Q - 3 (a) Write a java UDP socket program to transfer file from the server to client. The client will send the name of the file to server. The server will send contents of the file to the client. The client will reconstruct the file at its end. **[07]**

(b) Write the following stored procedures related to students' information (Table Student_Info): **[07]**

Get_Student_Name: IN Student_ID, OUT Student_Name

Update_Student_Mobile: IN Student_ID

Write a jdbc program to call the above mentioned stored procedure.

SECTION – II

Q - 4 Answer the following questions:

[07]

a) Which http method send by browser that gives the server what user data typed in the form?

A – post

B – get

C – put

D - forward

b) What is the difference between servlets and applets?

i. Servlets execute on Server; Applets execute on browser

ii. Servlets have no GUI; Applet has GUI

iii. Servlets creates static web pages; Applets creates dynamic web pages

iv. Servlets can handle only a single request; Applet can handle multiple requests

A - i, ii, iii are correct

B - i, ii are correct

C - i, iii are correct

D - i, ii, iii, iv are correct

c) Which are the session tracking techniques?

i. URL rewriting

ii. Using session object

iii. Using response object

iv. Using hidden fields

v. Using cookies

vi. Using servlet object

A - i, ii, iii, vi

B - i, ii, iv, v

C - i, vi, iii, v

D - i, ii, iii, v

d) Which of the following is not a core interface of Hibernate?

A - Configuration

B - Criteria

C - SessionManagement

D - Session

e) In which file database table configuration is stored?

A - .dbm

B - .hbm

C - .ora

D - .sql

f) Which of these methods are member of Remote class?

A - checkIP()

B - addLocation()

C - AddServer()

D - None of the mentioned

g) Which class acts as IoC Container?

A - ServletContext

B - DispatcherServlet

C - ApplicationContext

D - None of the mentioned

Q - 5 (a) Write a java servlet application having following requirement: **[07]**

- a) The application should restrict all the request coming from 13.12.12.11 IP address (Use filter).
- b) The main servlet will display the all the data related to HTTP header.
- c) The server will read the context and config parameter which are set in web.xml.

(b) Write an RMI application having following functionalities: **[07]**

- a) Client will send two numbers and the operation over the numbers to the server
- b) The server will perform the operation as per the client request and send back the result to the client. If the inputs or operation are not valid then server should send the message "Invalid Input"

OR

Q - 5 (a) What is servlet? Explain the lifecycle of the servlet. How HttpSession is used to maintain the user's session? **[07]**

(b) What is RMI registry? How stub and skeleton are generated? What is the default port of RMI? How Serializable interface is useful in RMI? **[07]**

Q – 6 *Explain the following in detail:* **[14]**

- a) Hibernate Framework
- b) Spring Framework